Assignment02 作业报告

1.读取文件代码

```
met = pd.read_csv(r"D:\python class\ESE5023_Assignments_12632710\Assignment 02\earthquakes-2026-10-14_20-32-06_+0800.tsv",sep="\t")

met.head(500)
met.info()
met.columns

met.head()
met.head(10)

met.tail()
met.tail(10)
Sig_Eqs = met.dropna(subset=["Id"]).copy()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6698 entries, 0 to 6697
Data columns (total 49 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Search Parameters                 1 non-null      str
 1   Id                                6697 non-null   float64
 2   Year                              6697 non-null   float64
 3   Mo                                6293 non-null   float64
 4   Dy                                6122 non-null   float64
 5   Hr                                4580 non-null   float64
 6   Mn                                4357 non-null   float64
 7   Sec                               3864 non-null   float64
 8   Tsu                               2124 non-null   float64
 9   Vol                               82 non-null     float64
 10  Country                           6697 non-null   str
 11  Area                              335 non-null    str
 12  Region                            6697 non-null   float64
 13  Location Name                     6697 non-null   str
 14  Latitude                          6634 non-null   float64
 15  Longitude                         6634 non-null   float64
 16  Focal Depth (km)                  3640 non-null   float64
 17  Mag                               4910 non-null   float64
 18  Mw                                1937 non-null   float64
 19  Ms                                3102 non-null   float64
 20  Mb                                1891 non-null   float64
 21  Ml                                224 non-null    float64
 22  Mfa                               14 non-null     float64
 23  Unk                               840 non-null    float64
 24  MMI Int                           3443 non-null   float64
 25  Deaths                            2272 non-null   float64
 26  Death Description                 2792 non-null   float64
 27  Missing                           27 non-null     float64
 28  Missing Description               29 non-null     float64
 29  Injuries                          1485 non-null   float64
 30  Injuries Description              1744 non-null   float64
 31  Damage ($Mil)                     657 non-null    float64
 32  Damage Description                4907 non-null   float64
 33  Houses Destroyed                  956 non-null    float64
 34  Houses Destroyed Description      2056 non-null   float64
 35  Houses Damaged                    651 non-null    float64
 36  Houses Damaged Description        1258 non-null   float64
 37  Total Deaths                      2144 non-null   float64
 38  Total Death Description           2582 non-null   float64
 39  Total Missing                     30 non-null     float64
 40  Total Missing Description         35 non-null     float64
 41  Total Injuries                    1510 non-null   float64
 42  Total Injuries Description        1774 non-null   float64
 43  Total Damage ($Mil)               647 non-null    float64
 44  Total Damage Description          4182 non-null   float64
 45  Total Houses Destroyed            991 non-null    float64
 46  Total Houses Destroyed Description  2141 non-null   float64
 47  Total Houses Damaged              600 non-null    float64
 48  Total Houses Damaged Description  1168 non-null   float64
dtypes: float64(45), str(4)
memory usage: 2.7 MB
```

1.1 各国或地区的地震死亡人数

我按照 Country 分组，对 Deaths 列求和，再按死亡人数从高到低排序，列出前 20 名。Deaths 表示地震直接死亡人数；Total Deaths 属于另一统计口径，可能包括相关次生灾害造成的死亡，因此没有将两列相加。缺失人数仍视为未知，不能理解为零死亡，表中的数值是已有记录之和。

```
•[3]: #1.1
deaths = Sig_Eqs.groupby("Country")["Deaths"].sum(min_count=1)
print(deaths.sort_values(ascending=False).head(20))
```

```
Country
CHINA           2139209.0
TURKEY          1199720.0
IRAN            1014453.0
ITALY            492236.0
SYRIA            419826.0
HAITI            323484.0
AZERBAIJAN       319251.0
JAPAN            242516.0
ARMENIA          177890.0
PAKISTAN         145085.0
IRAQ             136200.0
ECUADOR          135496.0
TURKMENISTAN     117412.0
PERU             102309.0
ISRAEL            93388.0
PORTUGAL          83572.0
GREECE            81162.0
CHILE             64277.0
INDIA             63507.0
TAIWAN            57154.0
Name: Deaths, dtype: float64
```

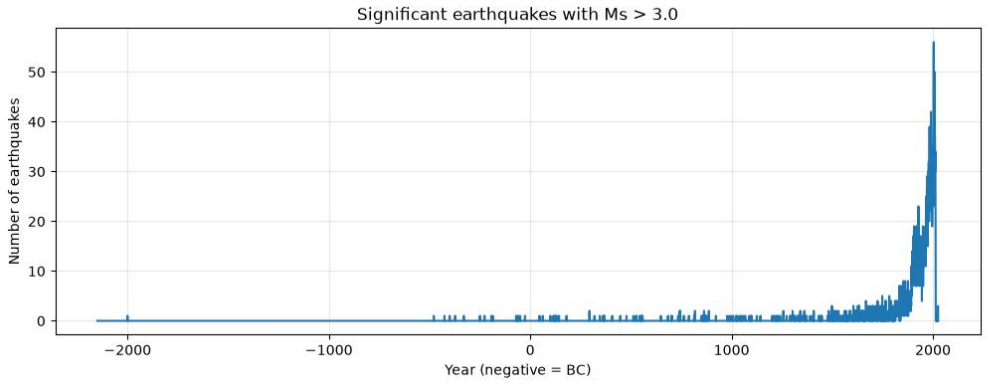
1.2 代码及结果

曲线显示近现代的地震记录明显多于古代，但不能据此断言实际地震频率上升。历史文献的完整程度、观测技术和报告能力都会影响收录。

```
#1.2
counts = Sig_Eqs[Sig_Eqs["Ms"] > 3.0].groupby("Year").size()

years = range(int(Sig_Eqs["Year"].min()), int(Sig_Eqs["Year"].max()) + 1)
counts = counts.reindex(years, fill_value=0).drop(0)

counts.plot(figsize=(12, 4))
plt.xlabel("Year (negative = BC)")
plt.ylabel("Number of earthquakes")
plt.title("Significant earthquakes with Ms > 3.0")
plt.grid(alpha=0.3)
plt.show()
```



1.3 代码及结果

我用主震级 Mag 的最大值定义“最大地震”。函数统计该国家或地区的全部记录数，包括 Mag 缺失的记录；若多个事件并列最大，则全部保留。若该组震级全部缺失，就无法确定最大地震。附录 A 按地震次数降序列出了全部 162 个国家或地区的结果。

```python
#1.3
def CountEq_LargestEq(country):
    data = Sig_Eqs[Sig_Eqs["Country"] == country]
    largest = data[data["Mag"] == data["Mag"].max()]

    return len(data), largest[["Year", "Mo", "Dy", "Location Name", "Mag"]]


countries = Sig_Eqs["Country"].value_counts()

for country in countries.index:
    count, largest = CountEq_LargestEq(country)
    print(f"\n{country}: {count} earthquakes")
    print(largest.to_string(index=False))
```

```
CHINA: 626 earthquakes
  Year  Mo   Dy            Location Name  Mag
1668.0 7.0 25.0 CHINA:  SHANDONG PROVINCE  8.5

JAPAN: 429 earthquakes
  Year  Mo   Dy  Location Name  Mag
2011.0 3.0 11.0 JAPAN:  HONSHU  9.1

INDONESIA: 427 earthquakes
  Year   Mo   Dy                          Location Name  Mag
2004.0 12.0 26.0 INDONESIA:  SUMATRA:  ACEH:  OFF WEST COAST  9.1

IRAN: 385 earthquakes
 Year   Mo   Dy         Location Name  Mag
856.0 12.0 22.0 IRAN:  DAMGHAN, QUMIS  7.9

TURKEY: 359 earthquakes
  Year   Mo   Dy                Location Name  Mag
1939.0 12.0 26.0            TURKEY:  ERZINCAN  7.8
2023.0  2.0  6.0 TURKEY: KAHRAMANMARAS; SYRIA  7.8

ITALY: 338 earthquakes
  Year  Mo   Dy                      Location Name  Mag
1915.0 1.0 13.0 ITALY:  MARSICA, AVEZZANO, ABRUZZI  7.5

GREECE: 296 earthquakes
  Year  Mo   Dy            Location Name  Mag
 365.0 7.0 21.0 GREECE:  CRETE:  KNOSSOS  8.0
1303.0 8.0  8.0           GREECE:  CRETE  8.0

USA: 283 earthquakes
  Year  Mo   Dy Location Name  Mag
1964.0 3.0 28.0        ALASKA  9.2


PHILIPPINES: 233 earthquakes
  Year  Mo   Dy                                  Location Name  Mag
1897.0 9.0 21.0 PHILIPPINES:  MINDANAO, ZAMBOANGA, SULU, ISABELA  8.7

MEXICO: 217 earthquakes
  Year  Mo   Dy               Location Name  Mag
1787.0 3.0 28.0 MEXICO:  SAN MARCOS, OAXACA  8.6

CHILE: 200 earthquakes
  Year  Mo   Dy                  Location Name  Mag
1960.0 5.0 22.0 CHILE:  PUERTO MONTT, VALDIVIA  9.5

PERU: 195 earthquakes
  Year  Mo  Dy                    Location Name  Mag
1716.0 2.0 6.0 PERU:  PUEBLO DE TORATA IN TACNA  8.8

RUSSIA: 159 earthquakes
  Year   Mo  Dy                Location Name  Mag
```

```
PAPUA NEW GUINEA: 107 earthquakes
  Year  Mo  Dy                        Location Name  Mag
1919.0 5.0 6.0 PAPUA NEW GUINEA:  SOLOMON ISLANDS  8.2

TAIWAN: 103 earthquakes
  Year  Mo  Dy Location Name  Mag
1920.0 6.0 5.0        TAIWAN  8.2

INDIA: 102 earthquakes
  Year  Mo   Dy Location Name  Mag
1950.0 8.0 15.0   INDIA-CHINA  8.6

COLOMBIA: 83 earthquakes
  Year  Mo   Dy                                      Location Name  Mag
1826.0 6.0 18.0 COLOMBIA:  ENGATIVA,BOGOTA,RAMIRIQUI,UMBITA,TUNJA  8.2
```

```
AFGHANISTAN: 72 earthquakes
  Year  Mo  Dy           Location Name  Mag
1909.0 7.0 7.0 AFGHANISTAN:  HINDU-KUSH  8.1

NEW ZEALAND: 72 earthquakes
  Year  Mo   Dy                                Location Name  Mag
1826.0 NaN  NaN                        NEW ZEALAND:  FJORDLAND  8.0
1855.0 1.0 23.0 NEW ZEALAND:  WELLINGTON,WAIOURU,WANGANUI,OTAKI  8.0

ECUADOR: 70 earthquakes
  Year  Mo   Dy       Location Name  Mag
1906.0 1.0 31.0 ECUADOR:  OFF COAST  8.8

VENEZUELA: 65 earthquakes
  Year  Mo  Dy      Location Name  Mag
1530.0 9.0 1.0 VENEZUELA:  CUMANA  8.0


  ALGERIA: 58 earthquakes
    Year   Mo   Dy      Location Name  Mag
  1980.0 10.0 10.0 ALGERIA:  NORTHERN  7.1

  ALBANIA: 56 earthquakes
    Year  Mo   Dy                                        Location Name  Mag
  1893.0 6.0 14.0 ALBANIA: HIMARA, DHERMI, KUC, KUDHESI, VLORE, KANINA  7.5

  PAKISTAN: 55 earthquakes
    Year   Mo   Dy           Location Name  Mag
  1945.0 11.0 27.0 PAKISTAN:  MAKRAN COAST  8.0

  CROATIA: 53 earthquakes
    Year  Mo  Dy                              Location Name  Mag
  1667.0 4.0 6.0 BALKANS NW:  CROATIA: DUBROVNIK:  RAGUSA  7.2
```

```
GUATEMALA: 47 earthquakes
  Year  Mo  Dy            Location Name  Mag
1942.0 8.0 6.0 GUATEMALA:  NEAR S COAST  7.7

FRANCE: 43 earthquakes
  Year  Mo   Dy            Location Name  Mag
1817.0 3.0 11.0 FRANCE:  CHAMONIX VALLEY  8.0

MYANMAR (BURMA): 43 earthquakes
  Year  Mo   Dy                            Location Name  Mag
1912.0 5.0 23.0 MYANMAR (BURMA):  MANDALAY, MOGOK, MAYMYO  8.0

EL SALVADOR: 42 earthquakes
  Year  Mo   Dy Location Name  Mag
1776.0 5.0 30.0   EL SALVADOR  7.9
```

NICARAGUA: 40 earthquakes

Year	Mo	Dy	Location Name	Mag
1898.0	4.0	29.0	NICARAGUA: LEON, CHINANDEGA, MANAGUA	7.9

ARGENTINA: 39 earthquakes

Year	Mo	Dy	Location Name	Mag
1894.0	10.0	27.0	ARGENTINA: LA RIOJA, SAN JUAN, MENDOZA	7.5
1912.0	12.0	7.0	ARGENTINA: SANTIAGO DEL ESTERO PROVINCE	7.5
1916.0	6.0	21.0	ARGENTINA	7.5
1977.0	11.0	23.0	ARGENTINA: SAN JUAN PROVINCE: MENDOZA	7.5

USA TERRITORY: 39 earthquakes

Year	Mo	Dy	Location Name	Mag
1902.0	9.0	22.0	GUAM: AGANA	8.1

COSTA RICA: 37 earthquakes

Year	Mo	Dy	Location Name	Mag
1950.0	10.0	5.0	COSTA RICA-NICARAGUA	7.7

SPAIN: 35 earthquakes

Year	Mo	Dy	Location Name	Mag
881.0	5.0	26.0	SPAIN; ALGERIA; MOROCCO	7.5

SYRIA: 34 earthquakes

Year	Mo	Dy	Location Name	Mag
1202.0	5.0	20.0	SYRIA: SOUTHWESTERN	7.6

AUSTRALIA: 34 earthquakes

Year	Mo	Dy	Location Name	Mag
1989.0	5.0	23.0	MACQUARIE ISLAND: MACQUARIE STATION; NEW ZEALAND	8.2

SWITZERLAND: 31 earthquakes

Year	Mo	Dy	Location Name	Mag
1601.0	9.0	18.0	SWITZERLAND	6.2

PORTUGAL: 29 earthquakes

Year	Mo	Dy	Location Name	Mag
-60.0	NaN	NaN	PORTUGAL: CABO SAN VICENTE	8.5
1755.0	11.0	1.0	PORTUGAL: LISBON	8.5
1761.0	3.0	30.0	PORTUGAL: LISBON, PORTO	8.5

TAJIKISTAN: 29 earthquakes

Year	Mo	Dy	Location Name	Mag
1907.0	10.0	21.0	TAJIKISTAN: KARATAG	7.4
1949.0	7.0	10.0	TAJIKISTAN	7.4

AZORES (PORTUGAL): 28 earthquakes

Year	Mo	Dy	Location Name	Mag
1968.0	2.0	28.0	AZORES	7.6

NEW CALEDONIA: 28 earthquakes

Year	Mo	Dy	Location Name	Mag
1875.0	3.0	28.0	NEW CALEDONIA: LOYALTY ISLANDS: LIFOU ISLAND	8.1
1920.0	9.0	20.0	NEW CALEDONIA: LOYALTY ISLANDS	8.1

TONGA: 26 earthquakes

Year	Mo	Dy	Location Name	Mag
1919.0	4.0	30.0	TONGA ISLANDS	8.1

ISRAEL: 25 earthquakes

Year	Mo	Dy	Location Name	Mag
-31.0	9.0	2.0	ISRAEL: QUMRAN,ARIHA (JERICHO)	7.0
746.0	1.0	18.0	ISRAEL: TEVERYA (TIBERIAS), ARIHA (JERICHO)	7.0

IRAQ: 24 earthquakes

Year	Mo	Dy	Location Name	Mag
1864.0	12.0	2.0	IRAQ: ZURBATIYAH, BADRAH, TURSAQ, BAGHDAD	6.4

PANAMA: 24 earthquakes

Year	Mo	Dy	Location Name	Mag
1882.0	9.0	7.0	PANAMA: SAN BLAS ARCHIPELAGO	8.3

```
KERMADEC ISLANDS (NEW ZEALAND): 23 earthquakes
  Year   Mo   Dy                          Location Name  Mag
1986.0 10.0 20.0              KERMADEC ISLANDS:  RAOUL  8.1
2021.0  3.0  4.0 KERMADEC ISLANDS: SSE OF RAOUL ISLAND  8.1

SOUTH KOREA: 22 earthquakes
  Year   Mo   Dy                    Location Name  Mag
  27.0 12.0  NaN           SOUTH KOREA:  KWANGJU  7.0
  89.0  7.0  NaN           SOUTH KOREA:  KWANGJU  7.0
 664.0  9.0 12.0           SOUTH KOREA:  KYONGJU  7.0
1681.0  6.0 12.0 SOUTH KOREA:  YANGYANG, SANCHOK  7.0

SLOVENIA: 22 earthquakes
  Year  Mo   Dy                               Location Name  Mag
1511.0 3.0 26.0 BALKANS NW:  SLOVENIA:  IDRIJA,SKOFJA LOKA  6.5


MOROCCO: 21 earthquakes
  Year  Mo  Dy            Location Name  Mag
2023.0 9.0 8.0 MOROCCO: MARRAKECH, SAFI  6.8

HAITI: 21 earthquakes
  Year  Mo  Dy       Location Name  Mag
1842.0 5.0 7.0 HAITI:  CAP-HAITIEN  8.0

FIJI: 20 earthquakes
  Year  Mo   Dy Location Name  Mag
2018.0 8.0 19.0  FIJI ISLANDS  8.2

DOMINICAN REPUBLIC: 19 earthquakes
  Year  Mo  Dy                               Location Name  Mag
1946.0 8.0 4.0 DOMINICAN REPUBLIC:  NORTHEASTERN COAST  7.9
```

.....省略其余结果。

#2 代码及结果

我先按逗号拆分 TMP，将温度转为数值，排除缺失值编码+9999，仅保留质量标记为 1 或 5、通过全部质量检查的记录。同一小时有多条报告时，先计算小时平均，再计算月平均，以免报告较多的小时占更大权重。

```python
#2
import pandas as pd
import matplotlib.pyplot as pltSD

met = pd.read_csv(
    r"D:\python class\ESE5023_Assignments_12632710\Assignment 02\Baoan_Weather_1998_2022.csv",
    usecols=["DATE", "TMP"]
)
met["DATE"] = pd.to_datetime(met["DATE"])

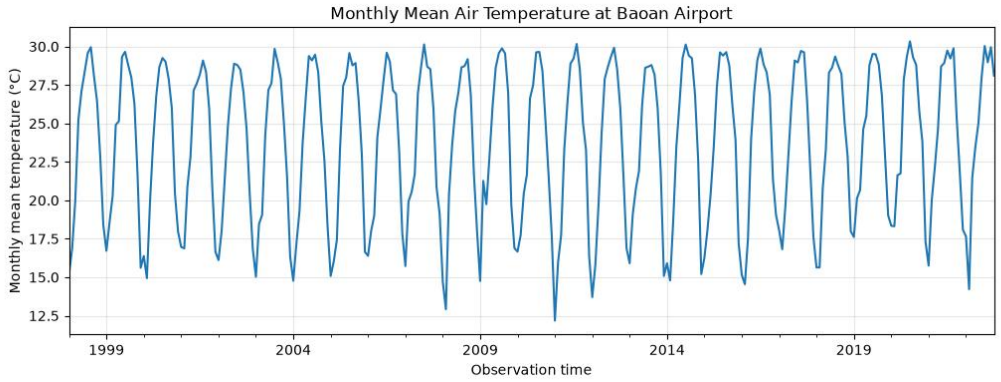
# 拆分温度编码和质量标记
tmp = met["TMP"].str.split(",", expand=True)
met["Temperature"] = pd.to_numeric(tmp[0], errors="coerce")

# 保留通过全部质量检查、且不是缺失值的温度
met = met[tmp[1].isin(["1", "5"]) & met["Temperature"].ne(9999)].copy()

# 转换为摄氏度
met["Temperature"] = met["Temperature"] / 10

# 先计算小时平均，再计算月平均
hourly = met.set_index("DATE")["Temperature"].resample("h").mean()
monthly = hourly.resample("MS").mean()

# 绘图
monthly.plot(figsize=(12, 4))
plt.xlabel("Observation time")
plt.ylabel("Monthly mean temperature (°C)")
plt.title("Monthly Mean Air Temperature at Baoan Airport")
plt.grid(alpha=0.3)
plt.show()
```



[10]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(
    r"D:\python class\ESE5023_Assignments_12632710\Assignment 02\ibtracs.ALL.list.v04r00.csv",
    usecols=range(17),
    skiprows=[1],
    parse_dates=["ISO_TIME"],
    skipinitialspace=True,
    keep_default_na=False,
    na_values=["", " ", "NOT_NAMED", "NAME"],
    low_memory=False
)

# 确保风速、经纬度为数字
for col in ["WMO_WIND", "LAT", "LON"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

df.head()
```

[10]:

	SID	SEASON	NUMBER	BASIN	SUBBASIN	NAME	ISO_TIME	NATURE	LAT	LON	WMO_WIND	WMO_PRES	WMO_AGENCY	TRACK_T
0	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 03:00:00	NR	10.9000	80.3000	NaN	NaN	NaN	
1	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 06:00:00	NR	10.8709	79.8265	NaN	NaN	NaN	
2	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 09:00:00	NR	10.8431	79.3524	NaN	NaN	NaN	
3	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 12:00:00	NR	10.8188	78.8772	NaN	NaN	NaN	
4	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 15:00:00	NR	10.8000	78.4000	NaN	NaN	NaN	

月平均气温的季节变化很明显，夏季高值接近 30 °C，冬季则明显偏低。为避免不完整的 2022 年影响年际比较，我对 1998—2021 年各年的 12 个月平均值再次取平均，并拟合直线，得到约 0.26 °C/10 年的正斜率。这提示宝安站在这一时段有一定升温倾向，但完整年份只有 24 年，且这里只分析了一个机场站点，不能直接据此代表整个深圳的长期气候变化。

TMP 列被拆分为温度值和质量标记。温度代码除以 10 得到摄氏度。标记为 +9999 的缺失值被排除，只保留质量标记为 1 或 5 的观测值，这表示所有质量检查都通过。首先计算每小时平均值，以合并一个小时内的多个报告，然后计算每月平均值。

每月平均温度显示出明显的季节性变化，夏天温度较高，冬天温度较低。在去除平均季节性循环后，估计的线性趋势约为每十年 +0.24° C，这表明宝安机场有升温的趋势。但统计显著性尚未检验，而且单个站点不一定能代表整个深圳。数据集截止到 2022 年 10 月 10 日，所以 2022 年 10 月的数据不完整，而 2022 年 11 月至 12 月的数据不可用。

#3

3.1 代码及结果

先按 SID 分组，再取每个风暴的最大 WMO_WIND。风速单位为节，排名第一的 Patricia 为 185 节，Allen 为 165 节，之后有多个风暴并列 160 节。不同机构采用的风速平均时长不同，跨海盆比较时需要考虑这一口径差异；同风速的排列顺序不代表强度有区别。

```
#3.1
storms = df.groupby("SID").agg(
    NAME=("NAME", "first"),
    WMO_WIND=("WMO_WIND", "max")
)

top10 = storms.nlargest(10, "WMO_WIND")
print(top10)
```

```
                  NAME  WMO_WIND
SID
2015293N13266  PATRICIA     185.0
1980214N11330     ALLEN     165.0
1935241N23291       NaN     160.0
1988253N12306   GILBERT     160.0
1997253N12255     LINDA     160.0
2005289N18282     WILMA     160.0
2019236N10314    DORIAN     160.0
1998295N12284     MITCH     155.0
2005261N21290      RITA     155.0
2009288N07267      RICK     155.0
```

3.2 代码及结果

```
#3.2
top20 = storms.nlargest(20, "WMO_WIND")
print(top20)
```

```
                  NAME  WMO_WIND
SID
2015293N13266  PATRICIA     185.0
1980214N11330     ALLEN     165.0
1935241N23291       NaN     160.0
1988253N12306   GILBERT     160.0
1997253N12255     LINDA     160.0
2005289N18282     WILMA     160.0
2019236N10314    DORIAN     160.0
1998295N12284     MITCH     155.0
2005261N21290      RITA     155.0
2009288N07267      RICK     155.0
2017242N16333      IRMA     155.0
1932304N15305       NaN     150.0
1955265N13307     JANET     150.0
1959249N19181     PATSY     150.0
1969226N18280   CAMILLE     150.0
1977242N27272     ANITA     150.0
1979238N12324     DAVID     150.0
1992230N11325    ANDREW     150.0
1994222N11267      JOHN     150.0
2005236N23285   KATRINA     150.0
```

3.3 代码及结果

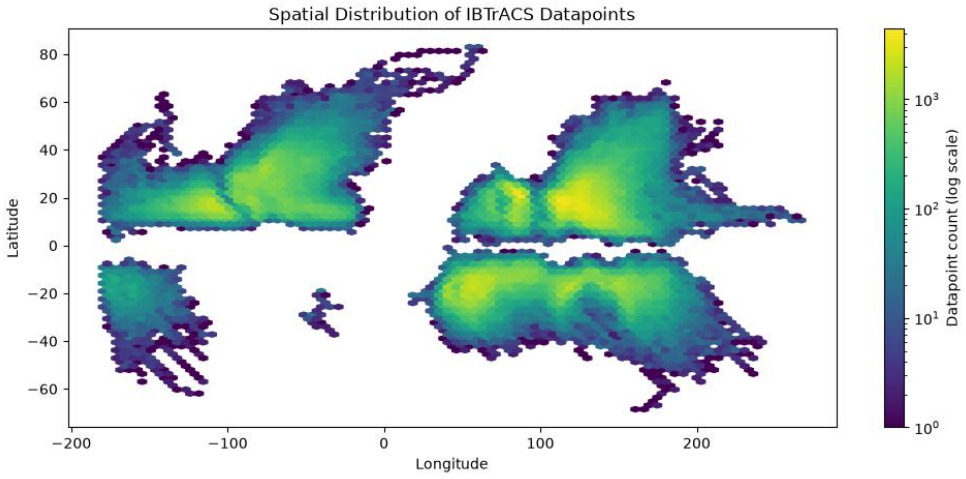
```python
#3.3
basin_counts = df["BASIN"].value_counts()
print(basin_counts)
```

```
BASIN
WP    236576
SI    160668
NA    124881
SP     67119
EP     62412
NI     55402
SA       119
Name: count, dtype: int64
```

3.4 代码及结果

```python
#3.4
positions = df.dropna(subset=["LAT", "LON"])

plt.figure(figsize=(12, 5))
plt.hexbin(
    positions["LON"], positions["LAT"],
    gridsize=80, bins="log", mincnt=1,
    cmap="viridis"
)
plt.colorbar(label="Datapoint count (log scale)")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Spatial Distribution of IBTrACS Datapoints")
plt.show()
```

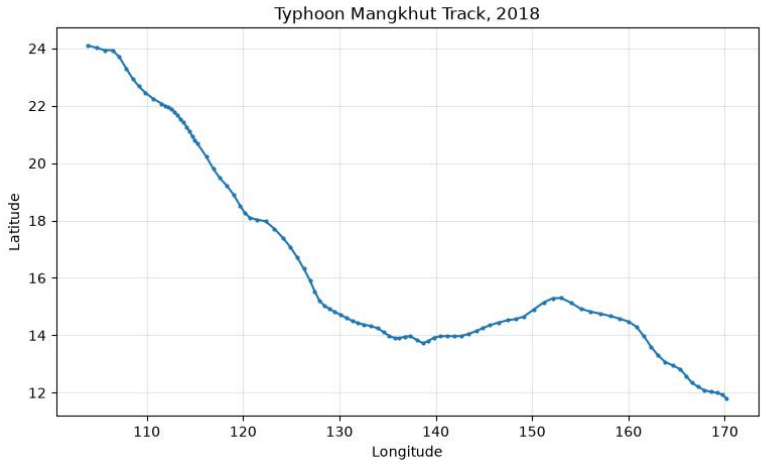


3.5 代码及结果

筛选 NAME 为 MANGKHUT 且 SEASON 为 2018 的记录，再按照 ISO_TIME 排序连接经纬度位置。这样既避免选入其他年份的同名台风，也保证路径按时间先后连接。

```
#3.5
mangkhut = df[
    (df["NAME"] == "MANGKHUT") &
    (df["SEASON"] == 2018)
].sort_values("ISO_TIME")

plt.figure(figsize=(9, 5))
plt.plot(mangkhut["LON"], mangkhut["LAT"], "-o", markersize=2)
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Typhoon Mangkhut Track, 2018")
plt.grid(alpha=0.3)
plt.show()
```



3.6 代码

```
#3.6
filtered = df[
    (df["ISO_TIME"] >= "1970-01-01") &
    df["BASIN"].isin(["WP", "EP"])
].copy()
```

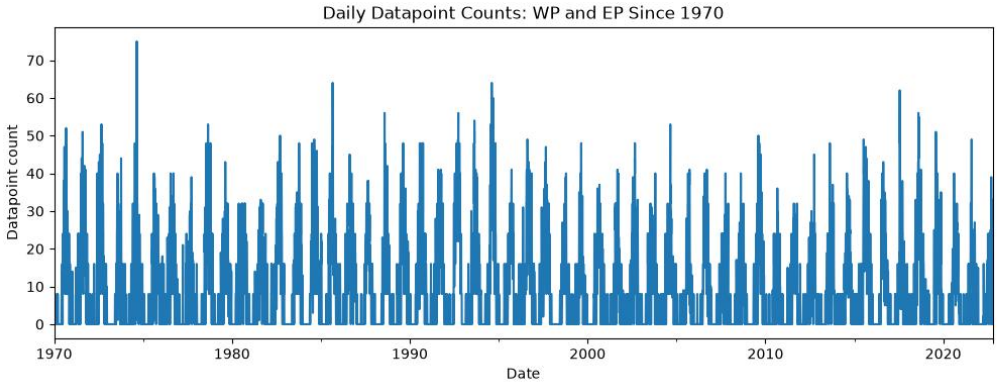
3.7 代码及结果

按天统计轨迹记录数，并补齐没有记录的日期。气候态的计算方法是把不同年份中相同年内日序的每日计数取平均，其中 1 月 1 日为第 1 天。曲线在太平洋气旋主要活动季节升高，其余季节较低。这个季节分布同时受到活跃风暴个数、持续时间和报告频率的影响。这里直接使用 dayofyear，闰年与非闰年 3 月以后的相同日序可能对应不同月日，第 366 天也只来自闰年。

```python
#3.7
daily = filtered.set_index("ISO_TIME").resample("D").size()

dates = pd.date_range(
    "1970-01-01",
    filtered["ISO_TIME"].max().normalize(),
    freq="D"
)
daily = daily.reindex(dates, fill_value=0)

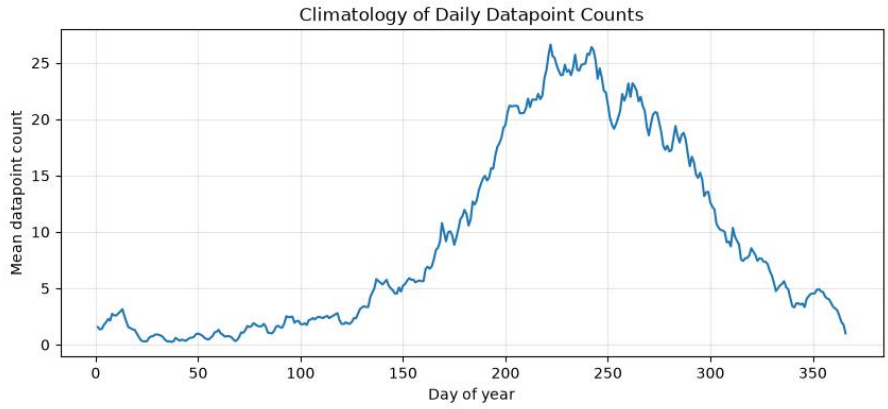
daily.plot(figsize=(12, 4))
plt.xlabel("Date")
plt.ylabel("Datapoint count")
plt.title("Daily Datapoint Counts: WP and EP Since 1970")
plt.show()
```



3.8 代码及结果

```python
#3.8
climatology = daily.groupby(daily.index.dayofyear).mean()

climatology.plot(figsize=(10, 4))
plt.xlabel("Day of year")
plt.ylabel("Mean datapoint count")
plt.title("Climatology of Daily Datapoint Counts")
plt.grid(alpha=0.3)
plt.show()
```

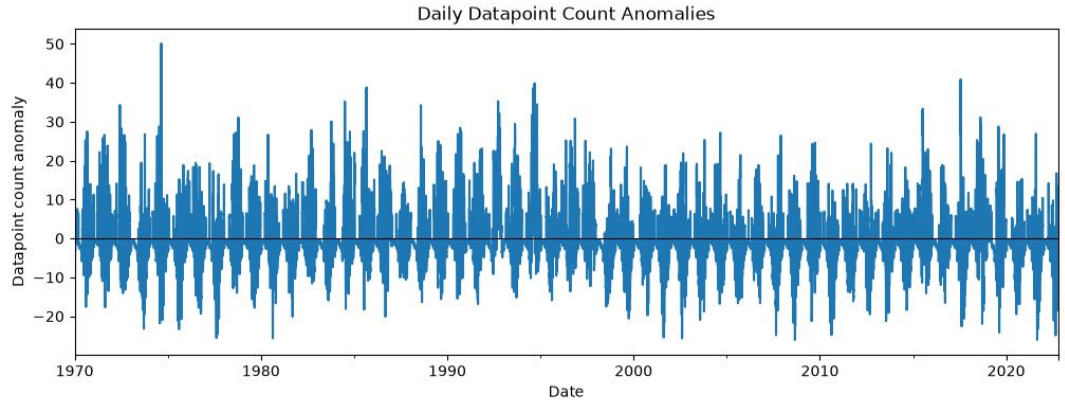


3.9 代码及结果

每日距平 = 当日记录点数 − 对应年内日序的气候态平均值。将每日距平按年取平均，就得到年平均距平；正值表示平均每天的记录数高于季节基准，负值表示低于基准。

```python
#3.9
expected = daily.index.dayofyear.map(climatology).to_numpy()
anomaly = daily - expected

anomaly.plot(figsize=(12, 4))
plt.axhline(0, color="black", linewidth=0.8)
plt.xlabel("Date")
plt.ylabel("Datapoint count anomaly")
plt.title("Daily Datapoint Count Anomalies")
plt.show()
```



3.10 代码及结果

```python
#3.10
# 计算年度平均异常
annual_anomaly = anomaly.resample("YS").mean()

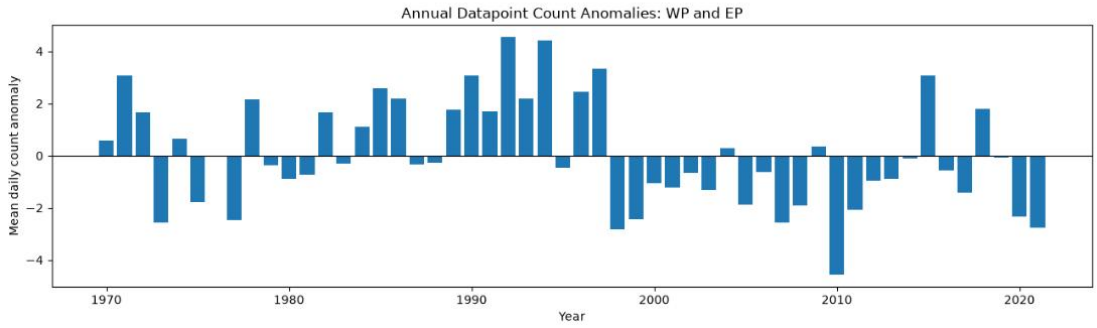
# 只比较完整年份，排除不完整的2022年
complete_years = annual_anomaly.loc["1970":"2021"].copy()

# 将日期索引转为整数年份，避免绘图时的频率转换问题
complete_years.index = complete_years.index.year

# 使用 matplotlib 绘制柱状图
plt.figure(figsize=(13, 4))
plt.bar(complete_years.index, complete_years.values)
plt.axhline(0, color="black", linewidth=0.8)
plt.xlabel("Year")
plt.ylabel("Mean daily count anomaly")
plt.title("Annual Datapoint Count Anomalies: WP and EP")
plt.tight_layout()
plt.show()

print("Highest positive anomalies:")
print(complete_years.nlargest(5).round(3))

print("\nLowest negative anomalies:")
print(complete_years.nsmallest(5).round(3))
```



1992 年和 1994 年的正距平最突出，分别约为+4.54 和+4.42 个记录点/日；2010 年的负距平最大，约为−4.57 个记录点/日，之后是 1998 年的−2.83 个记录点/日。因此这些年份的轨迹记录活动相对于季节基准较为异常。不过，这个指标不能直接说明当年生成了多少个风暴，也不能表示风暴强度。

```
Highest positive anomalies:
1992    4.540
1994    4.419
1997    3.334
1990    3.076
2015    3.071
dtype: float64

Lowest negative anomalies:
2010   -4.568
1998   -2.828
2021   -2.748
2007   -2.559
1973   -2.554
dtype: float64
```

4.代码及结果

选择了 NCEI Climate at a Glance 提供的全球陆地与海洋年平均温度距平数据。CSV 包含 1850—2025 年每年的一个温度距平，单位为℃，基准期为 1901—2000 年。负值表示低于基准期平均温度，是正常观测结果，不能作为坏值删除。

#4.1

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
#4.1
data = pd.read_csv(r"D:\python class\ESE5023_Assignments_12632710\Assignment 02\global_temperature_1850_2025.csv",
    comment="#",
    na_values=[-999, -999.9, -999.99, -9999]
)

data.columns = ["Year", "Anomaly"]

# 转为数字，无法转换的内容设为缺失值
for col in data.columns:
    data[col] = pd.to_numeric(data[col], errors="coerce")

print("Missing values:")
print(data.isna().sum())

# 删除缺失值和重复记录，按年份排序
data = data.dropna().drop_duplicates().sort_values("Year")
data["Year"] = data["Year"].astype(int)

print(data.head())
```

```
Missing values:
Year       0
Anomaly    0
dtype: int64
   Year  Anomaly
0  1850    -0.15
1  1851    -0.06
2  1852    -0.03
3  1853    -0.07
4  1854    -0.05
```

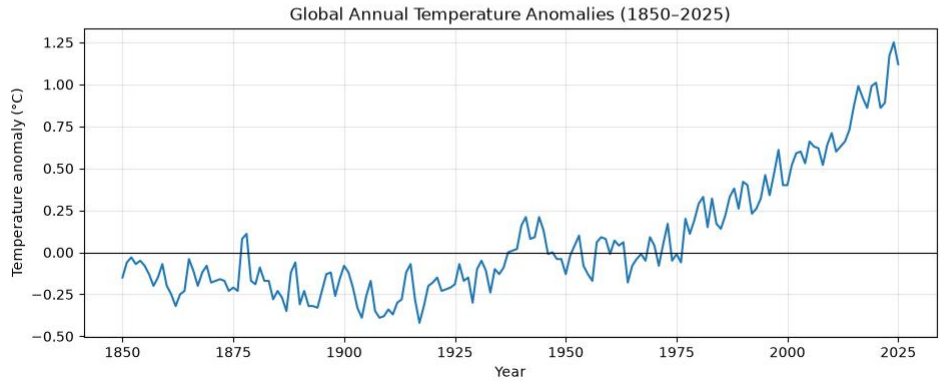
4.2 代码及结果

```
#4.2
plt.figure(figsize=(11, 4))

plt.plot(data["Year"], data["Anomaly"])
plt.axhline(0, color="black", linewidth=0.8)

plt.xlabel("Year")
plt.ylabel("Temperature anomaly (°C)")
plt.title("Global Annual Temperature Anomalies (1850–2025)")
plt.grid(alpha=0.3)

plt.show()
```



4.3 代码及结果

```
#4.3
# 样本数、均值、标准差、最小值、分位数、最大值
print(data["Anomaly"].describe().round(3))

# 最大值对应年份
print("\nWarmest year:")
print(data.loc[data["Anomaly"].idxmax()])

# 最小值对应年份
print("\nColdest year:")
print(data.loc[data["Anomaly"].idxmin()])

# 线性变化趋势
slope, intercept = np.polyfit(data["Year"], data["Anomaly"], 1)

print("\nTrend (°C per decade):", round(slope * 10, 3))
```

```
count    176.000
mean       0.063
std        0.361
min       -0.420
25%       -0.180
50%       -0.060
75%        0.210
max        1.250
Name: Anomaly, dtype: float64

Warmest year:
Year       2024.00
Anomaly       1.25
Name: 174, dtype: float64

Coldest year:
Year       1917.00
Anomaly      -0.42
Name: 67, dtype: float64

Trend (°C per decade): 0.057
```

统计项目	结果
年度样本数	176
均值	0.063 °C
中位数	-0.060 °C
样本标准差	0.361 °C
最小值	-0.42 °C (1917)
最大值	1.25 °C (2024)
第 25 / 75 百分位数	-0.18 / 0.21 °C
全时段线性斜率	0.057 °C /10 年

该数据集包含 NOAA NCEI 提供的 1850–2025 年全球陆地和海洋年温度异常，相对于 1901–2000 年的基准值。注释行已被跳过，数值字段已检查，缺失值和重复记录已删除。下载的数据中未发现缺失值。负异常值仍保留，因为它们表示低于基准的温度，而非无效观测值。该数据集包含 176 条年度观测记录。平均异常值为 0.063°C，中位数为 −0.060°C，样本标准差为 0.361°C。最低异常值出现在 1917 年，为 −0.42°C，最高异常值出现在 2024 年，为 1.25°C。时间序列表现出长期升温的趋势，同时每年都有波动。对 1850–2025 年进行线性拟合得到的趋势大约是每十年 +0.057° C。这概括了整个时期，并不意味着升温速率是恒定的。未进行统计显著性检验。

总结：

这次作业中，四类数据都能画出时间变化，但这些曲线能支持的结论并不相同。地震数据库中的近现代记录更多，但历史收录和 Ms 缺失限制了对真实地震频率的判断。宝安机场气温有明显季节性，在完整年份内呈一定升温倾向。台风轨迹计数在 1992、1994 年偏多、2010 年偏少，但计数也受到风暴持续时间和报告频率影响。全球年平均温度距平则在 1850—2025 年的长时段内总体上升。

具体操作中，最容易影响结果的是一些看似很小的细节：+9999 并非温度，NA 在风暴数据中是海盆标签，缺失值不能直接填成零，最后一年也未必完整。先把变量含义和筛选规则弄清楚，再讨论曲线变化，得到的结论才比较可靠。